

Non-Local Morphological PdEs for Image Processing

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Abstract

Image processing has multitude of applications ranging from medical imaging to space research to social media. Building on the foundational work by Elmoataz et. al. on non-local morphological Partial Differential Equations (PDEs) and p-Laplacian operators, this project explores their application in enhancing image processing tasks.

We begin with brief introduction to image processing and the traditional methods for denoising and morphology. We provide the necessary mathematical background. We explore the theoretical framework of image processing using this new class of non-local discrete anisotropic p-Laplacian interpolating between normalized Laplacian and infinity Laplacian. We also demonstrate the implementations of these ideas for image denoising and smoothing over a range of parameter and graph configurations to showcase the advantages of this over local methods.

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Chapter 1

Introduction

1.1 Introduction

Image processing encompasses a range of techniques aimed at enhancing, analyzing, and manipulating images for a variety of applications. One of the key challenges in image processing is denoising, which involves the removal of noise while preserving image details. Noise in images can arise from various sources, such as sensor imperfections, transmission errors, etc. The process of denoising requires a thorough understanding of how images are stored and processed. Digital images are represented as a grid of pixels, each pixel containing color information expressed in terms of intensity values for different color channels. We will primarily work with gray-scale images in this project. Manipulating these pixel values according to mathematical models allows for alteration of image attributes. Morphological PDEs are one such model which delivers better results than traditional simpler models as they're better at preserving important features. But before discussing that, we'll go over the traditional methods to get an overview of what are the fundamental ideas in image processing.

1.2 Traditional methods

Traditional image processing methods, serve as basis for more advanced PDE based methods for image denoising and smoothing operations.

- **Mean Filters:** This filter works by replacing the value of each pixel with the average of it's neighbors within a specific window. This effectively blurs the image, smoothing

out the sudden variations caused by noise since noise is generally assumed to be random and averaging it over a region diminishes its effect. But this method also blurs across the edges and the details are lost. (See Figure 1.1b)

- **Median Filters:** This filter works by replacing the value of each pixel with the median value of the neighboring pixels. This effectively blurs the image without blurring across the major edges. This prevents the loss of edges better than mean filters (specifically in the case of salt and pepper noise) in the image but a lot of the details are still lost. (See Figure 1.1c)



(a) Noisy Image (b) Mean Filter (c) Median Filter

Figure 1.1: Denoising results of traditional filters.

- **Dilation and Erosion:**
  - *Structuring Element* is a specific shape (like a square or circle) that determines how pixels are added or removed from an object's boundary.
  - *Dilation* is the process of adding pixels to boundaries of objects in an image. When the structuring element is centered on a pixel, if any part of it overlaps with the object, the pixel under the center is added to the object. For binary images, this is same as replacing every pixel's value by the maximum pixel value in the region covered by a structuring element centered on it.
  - *Erosion* is the process of removing pixels from boundaries of objects in an image. When the structuring element is centered on a pixel, if it does not completely fit inside the object's boundary, the pixel under the center is removed from the object. For binary images, this is same as replacing every pixel's value by the minimum pixel value in the region covered by a structuring element centered on it.

Chapter 2

PdEs on Graphs

2.1 Essential Definitions and Notations

2.1.1 Notations

- Any discrete domain can be viewed as a weighted graph $G = (V, E, w)$ consisting of a finite set V of N vertices and a finite set $E \subset V \times V$ of edges, with a weight function $w : V \times V \to [0, 1]$. The edge set is defined as $E \coloneqq \{(u, v) | w(u, v) \neq 0\}$. Two vertices u and v are said to be adjacent, denoted as $u \sim v$, if $(u, v) \in E$. The degree of a vertex u, denoted as $\delta_u(u) = \sum_{v \sim u} w(u, v)$. The neighborhood of a vertex u is the set of vertices adjacent to u, denoted $N(u)$. For a connected set of vertices $A \subset V$, the boundary set of A, $\partial A = \{u \in A^c : \exists v \in A, (u, v) \in E\}$.

- The Hilbert space of real valued functions on the vertices of the graph is denoted $\mathcal{H}(V)$, and each function $f : V \to \mathbb{R}$ in $\mathcal{H}(V)$ assigns a real value $f(u)$ to each vertex $u \in V$. Similarly, let $\mathcal{H}(E)$ be the Hilbert space of real valued functions defined on the edges of the graph, endowed with the inner products: $\langle f, g \rangle_{\mathcal{H}(V)} = \sum_{u \in V} f(u)g(u)$ and $\langle F, G \rangle_{\mathcal{H}(E)} = \sum_{u \in V} \sum_{v \in E} F(u, v)G(u, v)$, for all $f, g \in \mathcal{H}(V)$ and $F, G \in \mathcal{H}(E)$.

- Given a function $f : V \to \mathbb{R}$, the integral of f is defined as $\int_V f = \sum_{u \in V} f(u)$.

- The L_p norm of a function f is given by $\|f\|_{L_p} = \left(\sum_{u \in V} |f(u)|^p \right)^{\frac{1}{p}}$ for $1 \leq p < \infty$, and the L_∞ norm is given by $\|f\|_\infty = \max_{u \in V} |f(u)|$.

- We denote $max(x, 0)$ as $(x)^+$ and $-min(0, x)$ as $(x)^-$.

2.1.2 Operators on Graphs

We need to define various, in general, continuous operators in the context of weighted graphs to be able to adapt continuous mathematical models to discrete graphs. And we need to have a notion of norms on these operators to work with regularization functions on weighted graphs.

- **Gradient Operators:**
  - The non-local gradient (or *difference operator*) of a function $f \in \mathcal{H}(V)$, denoted G_w, is defined as $(G_w f)(u,v) \coloneqq \gamma(w(u,v)(f(u)-f(v))$. We will denote $\gamma(w(u,v))$ as γ_{uv} from now on. From here onwards, we consider $\gamma(w(u,v)) = \sqrt{w(u,v)}$ unless stated otherwise.
  - k-order directional derivative of a function $f \in \mathcal{H}(V)$ is $\partial_v^k f(u) = (-1)^{k+1}\gamma_{uv}^k(f(v)-f(u))$. And $(\partial_v^{\pm} f)(u) = ((\partial_v f)(u))^{\pm}$.
  - Discrete non-local weighted gradient of a function $f \in \mathcal{H}(V)$, denoted by $\nabla_w f$, is defined on a vertex $u \in V$ as the vector of all partial derivatives, $(\nabla_w f)(u) \coloneqq ((\partial_v f)(u))_{v\in V}^T$. And $(\nabla_w^{\pm} f)(u) = ((\partial_v^{\pm}) f(u))_{v\in V}^T$.
- **Divergence Operators:**
  - Adjoint of *difference operator*, denoted by G_W^*, of a function $H \in \mathcal{H}(E)$, is given by: $(G_w^* H)(u) \coloneqq \sum_{v\sim u} \gamma_{uv}(H(v,u) - H(u,v))$. The *divergence operator* defined as $D_w = -G_w^*$, measures the net outflow at each vertex of a function in $\mathcal{H}(E)$.
  - $\sum_{u\in V}\sum_{v\in V} G_w f(u,v) = 0$, $f \in \mathcal{H}(V)$ and $\sum_{u\in V} D_w F(u) = 0$, $F \in \mathcal{H}(E)$.
- **p-Laplacian:**
  - The discrete version of anisotropic p-Laplacian operator of a function $f \in \mathcal{H}(V)$, denoted as $\Delta_{w,p} : \mathcal{H}(V) \to \mathcal{H}(V)$ has been computed as:

$$\left(\Delta_{w,p} f\right)(u) = \sum_{v\sim u} \gamma_{uv}^p |f(v)-f(u)|^{p-2}(f(v)-f(u)) \tag{2.1}$$

  - This operator is non-linear for $p \neq 2$. For $p = 2$, this is just the Laplacian given by:

$$\left(\Delta_{w,2} f\right)(u) = \sum_{v\sim u} w(u,v)(f(v)-f(u)) \tag{2.2}$$

- **Norms on Gradient Operators:**
  - $\left\|(\nabla_w^{\pm} f)(u)\right\|_p = \left[\sum_{v \in V} \gamma_{uv}^p \left((f(v) - f(u))^{\pm}\right)^p\right]^{\frac{1}{p}}$
  - $\left\|(\nabla_w^{\pm} f)(u)\right\|_\infty = \max_{v \in V} \left(\gamma_{uv}(f(v) - f(u))^{\pm}\right)$

2.2 Graph Construction

We can transform a discrete data (like an image) to a weighted graph structure with vertices corresponding to data points and edges modeling the relationship between data points. The edge weights are configured based on spatial distance and/or using a pairwise feature distance function $\mu : V \times V \to \mathbb{R}^+$.

Grid Graphs are the most natural structures to describe an image with a graph. Each pixel is connect by an edge to its adjacent pixels. The size of neighborhood (extent of adjacency) can be varied depending upon the use case. Classical graphs are 4-adjacency and 8-adjacency. Larger adjacency is used to obtain non-local graphs. Depending on the particular image processing task and the particular image involved some other graphs such as k-NN graphs, region adjacency graphs, triangulated mesh graphs, etc,. can be more suitable.

In grid graphs, every vertex corresponds to a pixel in the image, and $f(u)$ stores the color information at the pixel corresponding to vertex u(pixel intensity in the case of gray scale images). To define the distance function μ, we often consider patches of pixels centered at the target pixels (3×3 window, for instance) and compute the distance between these patches(using euclidean distance, for instance). This assists in capturing the local structure involving the pixels. We can now use this distance function to define the similarity and weight functions. The similarity function $g : E \to \mathbb{R}^+$ satisfies:

$$w(u,v) = \begin{cases} g(u,v) & \text{if } (u,v) \in E \\ 0 & \text{otherwise} \end{cases}$$

Usual similarity functions are as follows:

$$g_0(u,v) = 1$$
$$g_1(u,v) = e^{\left(-\mu(u,v)/\sigma^2\right)} \text{ with } \sigma > 0$$

where σ depends on the variation of μ and can be used to control the extent of effect of similarity.

2.3 Non-Local Morphological PdEs on Graphs

We mimic the PDEs models on continuous domains with PdEs on graphs by replacing continuous (differential) spatial operators by discrete (difference) spatial operators on graphs as described in section 2.1. Let $f^0 \in \mathcal{H}(V)$ be a given function defined on the vertices of a weighted graph $G = (V, E, w)$. In the current context, this function represents the initial state of the image, and let $h \in \mathcal{H}(V)$ represents the state of clean image, such that $f^0 = h + n$, where $n \in \mathcal{H}(V)$ is the additive noise. In order to recover the clean function h, we try to find a function $f \in \mathcal{H}(V)$, which is regular enough on G, and also close enough to f^0. We can compute this by using Euler-Lagrange equation to minimise an energy functional which involves a regularization term plus an approximation term. We model this as following,

$$\begin{aligned} h &\approx \arg\min_{f:V\to\mathbb{R}} J_{w,p}(f) + \frac{\lambda}{2}\left\|f - f^0\right\|_2^2, \\ \text{where } J_{w,p}(f) &\overset{\text{def.}}{=} \frac{1}{2p}\sum_{u\in V}\left\|\left(\nabla_w f\right)(u)\right\|_p^p \end{aligned} \tag{2.3}$$

After applying Euler-Lagrange Equation to this we get a non-linear system, which after linearizing by Guass-Jacobi iterative algorithm (see [1] for details), gives:

$$\begin{cases} \beta_{uv}^{p,n} = \sum_{v\sim u}\gamma_{uv}^p\left|f^n(v) - f^n(u)\right|^{p-2}, & \forall (u,v)\in E, \\ f^{n+1}(u) = \frac{\lambda f^0(u) + \sum_{v\sim u}\beta_{uv}^{p,n} f^n(v)}{\lambda + \sum_{v\in V}\beta_{uv}^{p,n}}, & \forall u \in V. \end{cases} \tag{2.4}$$

The choice of regularization parameters and the choice of the graph leads to several well-known filters in the context of image smoothing and denoising. In particular for $p = 2$, $\lambda = 0$ (no fitting term) and $\gamma_{uv} = \sqrt{w(u,v)}$, we get the non-local mean filter: (NLM).

$$f^{n+1}(u) = NLM(f^n)(u) = \frac{\sum_{v\sim u} w(u,v) f^n(v)}{\sum_{v\sim u} w(u,v)} \tag{2.5}$$

For constant weighted graph $(w(u,v) = 1, \forall (u,v) \in E)$, this is analogous to traditional mean filter.

We also have PdEs discrete analogue of continuous PDEs-based dilation(δ) and erosion(ϵ) formulations over graphs as proposed in [2]. For a given initial function $f^0 \in \mathcal{H}(V)$, using the structuring sets $B = \{x : |x|_p \le 1\}$

$$\begin{aligned} \frac{\partial\delta(f)(u)}{\partial t} &= \partial_t f(u) = +\left\|\left(\nabla_w^+ f\right)(u)\right\|_p, \text{ and} \\ \frac{\partial\varepsilon(f)(u)}{\partial t} &= \partial_t f(u) = -\left\|\left(\nabla_w^- f\right)(u)\right\|_p \quad , \forall u \in V \end{aligned} \tag{2.6}$$

where f is the modified function with the initial condition $f = f^0$ at $t = 0$,

The dilation and erosion process in equation (2.6) can be interpreted as a family of non-local digital averaging filters. In particular for $p = \infty$, the dilation process can be expressed as the following iterative PdE:

$$f^{n+1}(u) = f^n(u) + \Delta t \max_{v \sim u} \left(\sqrt{w(u,v)} \max(f^n(v) - f^n(u), 0) \right) \tag{2.7}$$

In the case of $\Delta t = 1$, equation (2.7) can be interpreted as an iterative non-local dilation (NLD) given by:

$$NLD(f)(u) = f(u) + \max_{v \sim u} \left(\sqrt{w(u,v)} \max(f(v) - f(u), 0) \right) \tag{2.8}$$

Similarly, we can define an iterative non-local erosion (NLE) given by:

$$NLE(f)(u) = f(u) - \max_{v \sim u} \left(-\sqrt{w(u,v)} \min(f(v) - f(u), 0) \right) \tag{2.9}$$

For a constant weighted graph, $w(u,v) = 1, \forall (u,v) \in E$, and considering graph construction such that $u \in N(u), \forall u \in V$, NLD and NLE recover the true morphological formulations, as in this case for NLD we get:

$$f^{n+1}(u) = f^n(u) + \max_{v \sim u} \left(f^n(v) - f^n(u) \right) = \max_{v \sim u} f^n(v)$$

Similarly for NLE we get:

$$f^{n+1}(u) = f^n(u) + \max_{v \sim u} \left(f^n(u) - f^n(v) \right) = \min_{v \sim u} f^n(v)$$

These are analogous to traditional dilation and erosion processes.

Chapter 3

Non-local p-Laplacian for Filtering and Morphology

In this section we will introduce a new class of non-local p-Laplacian that interpolates between infinite Laplacian and Laplacian, and can serve as a combined framework for morphological and filtering operators on graphs.

3.1 Laplacian and ∞-Laplacian on graphs as Morphological Processes

First, we'll show this new relation to be used in expressing Laplacian and ∞-Laplacian as combination of morphological processes.

Proposition 1: For $1 \leq p < +\infty$, at a vertex $u \in V$

$$\left(\Delta_{w,(p+1)} f\right)(u) = \left\| \left(\nabla^{+}_{w'} f\right)(u) \right\|_p^p - \left\| \left(\nabla^{-}_{w'} f\right)(u) \right\|_p^p \tag{3.1}$$

where $w'(u,v) = w(u,v)^{(p+1)/p}$.

Proof: We have,

$$\left(\Delta_{w,p} f\right)(u) = \sum_{v \sim u} \gamma_{uv}^p |f(v) - f(u)|^{p-2}(f(v) - f(u))$$

Since $|x| = x^+ + x^-$ and $x = x^+ - x^-$, taking $A = (f(v) - f(u))$, we get:

$$\left(\Delta_{w,p+1} f\right)(u) = \sum_{v \sim u} \gamma_{uv}^{p+1} (A^+ + A^-)^{p-1}(A^+ - A^-)$$

Now, since $A^+A^- = 0$, expanding $(A^+ + A^-)^{p-1}$ we obtain equation (3.1). Substituting $p = 1$, we get:

$$\left(\Delta_{w,2} f\right)(u) = \left\| \left(\nabla^+_{w'} f\right)(u) \right\|_1 - \left\| \left(\nabla^-_{w'} f\right)(u) \right\|_1 \tag{3.2}$$

The Laplacian operator can therefore be expressed as the difference of two morphological external and internal gradients.

As in the continuous case as interpreted in [3], the ∞-Laplacian can be formally derived as minimization of the following energy on graphs, as p goes to infinity.

$$J_{w,p}(f) = \sum_{u \in V} \left\| (\nabla_w f)(u) \right\|_p^p \tag{3.3}$$

The minimization leads to p-Laplacian equation $\Delta_{w,p} f(u) = 0, \forall u \in V$. From equation (3.1), we have

$$\left(\Delta_{w,p} f\right)(u) = \left\| \left(\nabla^+_{w'} f\right)(u) \right\|_{p-1}^{p-1} - \left\| \left(\nabla^-_{w'} f\right)(u) \right\|_{p-1}^{p-1} = 0$$

with $w'(u,v) = w(u,v)^{p/(p-1)}$, which is simplified into

$$\left\| \left(\nabla^+_{w'} f\right)(u) \right\|_{p-1} - \left\| \left(\nabla^-_{w'} f\right)(u) \right\|_{p-1} = 0.$$

Taking $p \to \infty$,

$$\left\| \left(\nabla^+_{w'} f\right)(u) \right\|_{\infty} - \left\| \left(\nabla^-_{w'} f\right)(u) \right\|_{\infty} = 0.$$

Based on this, we propose a new definition of ∞-Laplacian for graphs:

$$\left(\Delta_{w,\infty} f\right)(u) = \frac{1}{2} \left[\left\| \left(\nabla^+_{w'} f\right)(u) \right\|_{\infty} - \left\| \left(\nabla^-_{w'} f\right)(u) \right\|_{\infty} \right] \tag{3.4}$$

This is fitting since for constant weight function $(w(u,v) = 1, \forall (u,v) \in E)$, this reduces to

$$\left(\Delta_{1,\infty} f\right)(u) = \frac{1}{2} \left[\max_{v_1 \sim u} f(v_1) + \min_{v_2 \sim u} f(v_2) \right] - f(u) \tag{3.5}$$

$\left(\Delta_{1,\infty} f\right)(u)$ in equation (3.5) is well-known finite difference approximation of infinity Laplacian (morphological Laplacian). Now we can rewrite $\left(\Delta_{w,\infty} f\right)(u)$ as:

$$\left(\Delta_{w,\infty} f\right)(u) = \frac{1}{2} \left[NLD(f)(u) + NLE(f)(u) \right] - f(u) \tag{3.6}$$

So, we have expressed ∞-Laplacian on graphs as combination of morphological processes.

3.2 Non-local Normalized p-Laplacian

We now propose a new family of p-Laplacian which will serve as a combined framework for various image processing tasks by interpolating between Laplacian and ∞-Laplacian. We do so by discretizing the game p-Laplacian as defined in [4]. Recall (see [5] for details) that in the continuous case

$$\Delta_p f = |\nabla f|^{p-4} \{(p-2)\Delta_\infty f + |\nabla f|^2 \Delta f\} \tag{3.7}$$

The normalized version of this, referred to as game p-Laplacian is defined as:

$$\begin{aligned} \Delta_p^N f &= \frac{1}{p|\nabla f|^{p-2}} \Delta_p f \\ &= \frac{p-2}{p} \Delta_\infty^N f + \frac{1}{p} \Delta f \\ &= \alpha \Delta_\infty^N f + \beta \Delta f \end{aligned} \tag{3.8}$$

where $\alpha = \frac{p-2}{p}$ and $\beta = \frac{1}{p}$ and $\Delta_\infty^N f = \frac{1}{|\nabla f|^2} \Delta_\infty f$. Discretizing equation (3.8) using our previously defined discrete Laplacian (equation (2.2)) and ∞-Laplacian (equation (3.4)), we propose a discrete non-local version of normalized p-Laplacian.

$$\Delta_{w,\alpha,\beta} f(u) \stackrel{\text{def}}{=} \alpha \left(\Delta_{w,\infty} f\right)(u) + \beta \left(\Delta_{w,2} f\right)(u) \tag{3.9}$$

Usual Laplacian and ∞-Laplacian can be obtained from this with different values of α and β. From here onwards, we'll refer to equation (3.9) as non-local p-Laplacian. Non-local p-Laplacian can be expressed in morphological operators as:

$$\Delta_{w,\alpha,\beta} f = \frac{\alpha}{2} \left[NLD(f) + NLE(f)\right] + \beta NLM(f) - f \tag{3.10}$$

3.3 Non-local p-Laplacian Diffusion on Graphs

In image processing, non-local models have shown advantages over traditional local models as they don't only rely on smoothing over spatially close pixels. Non-local models thus better preserve important features in the image like edges, and repetitive structures (textures). Now, we'll express the non-local diffusion problem involving non-local p-Laplacian as an iterative non-local average filtering process. Given the diffusion problem:

$$\begin{cases} \frac{\partial f(u,t)}{\partial t} = \Delta_{w,\alpha,\beta} f(u,t) \\ f(u, t=0) = f^0(u) \end{cases} \tag{3.11}$$

where $f : V \times [0,T] \to \mathbb{R}$ is a function and $f^0 : V \to \mathbb{R}$ is the initial value of f. This can be approximated with the scheme:

$$\frac{\partial f(u,t)}{\partial t} = \frac{f^{n+1}(u) - f^n(u)}{\Delta t} \tag{3.12}$$

where $f^n(u) = f(u, n\Delta t)$, $\Delta t = \lambda$. Then,

$$\begin{aligned} f^{n+1}(u) =& f^n(u) + \lambda\alpha\Delta_{w,\infty} f^n(u) + \lambda\beta\Delta_{w,2} f^n(u) \\ =& f^n(u) + \lambda\alpha \left[\frac{\left\| \left(\nabla_w^+ f\right)(u) \right\|_\infty - \left\| \left(\nabla_w^- f\right)(u) \right\|_\infty}{2} \right] + \lambda\beta \left[\frac{\sum w(u,v) f^n(v)}{\delta_w(u)} - f^n(u) \right] \\ =& (1-\lambda) f^n(u) + \lambda\alpha \left[\frac{NLE\left(f^n\right)(u) + NLD\left(f^n\right)(u)}{2} \right] + \lambda\beta NLM\left(f^n\right)(u). \end{aligned} \tag{3.13}$$

We can now use this iterative scheme in equation (3.13) with empirically fine tuned values of λ, with various combinations of α and β values to perform denoising and smoothing on images.

Chapter 4

Fractional Order Anisotropic Diffusion

4.1 Introduction

Fractional derivatives extend the concept of integer-order derivatives, allowing for more generalized operations in image processing. The flexibility and intrinsic "non-local" nature of fractional derivatives makes them particularly useful in capturing intricate structures within images which are often overlooked by integer-order differential equations. The fractional derivative can be interpreted through several definitions such as Riemann-Liouville and Grunwald-Letnikov, although in this work, the frequency domain (Fourier domain) definition is primarily used for practical computations.

4.2 Essential Definitions and Notations

- For any function $f(t) \in L^2(\mathbb{R})$, the Fourier transform of it is $\hat{f}(w) = \int_{\mathbb{R}} f(t)e^{-jwt}dt$. The first order derivative in the frequency domain is

$$Df(t) \overset{F}{\leftrightarrow} (jw)\hat{f}(w) \tag{4.1}$$

  So the αth-order derivative, $D_\alpha f(t) \overset{F}{\leftrightarrow} (jw)^\alpha \hat{f}(w)$. Now we can extend α beyond integers, to get the definition of fractional order derivative in the frequency domain

$$D_\alpha f(t) \overset{F}{\leftrightarrow} (jw)^\alpha \hat{f}(w), \quad \alpha \in \mathbb{R}^+$$

- For any $g(x,y) \in L^2(\mathbb{R}^2)$, the corresponding 2-D Fourier transform of it is

$$\hat{g}(w_1, w_2) = \int_{\mathbb{R}^2} g(x,y) \exp(-j(w_1 x + w_2 y))\, dxdy. \tag{4.2}$$

  Thus, fractional-order partial derivatives are

$$D_{\alpha x} g = F^{-1}\left((jw_1)^{\alpha} \hat{g}(w_1, w_2)\right) \tag{4.3}$$

$$D_{\alpha y} g = F^{-1}\left((jw_2)^{\alpha} \hat{g}(w_1, w_2)\right) \tag{4.4}$$

  where F^{-1} is the inverse 2-D continuous Fourier transform operator.

- Since images have bounded domains and Fourier transforms are defined for functions given on the whole space, we will consider the prolongation of the image function from Ω onto $\mathbb{R}^2$. The Sobolev space $H^s(\mathbb{R}^2)$ is a Hilbert space equipped with the inner product.

$$\langle g, h \rangle_s = \int \left(1 + |w_1|^2 + |w_2|^2\right)^s \hat{g}\bar{\hat{h}} dw_1 dw_2$$

  and the associated norm is given by $\|g\|_s = \sqrt{\langle g, h \rangle_s}$. The space $H^s(\Omega)$ is defined as the restriction of $H^s(\mathbb{R}^2)$ on Ω. Under the assumption that Ω has certain regularities, then the prolongation theorem (see [6] for details) ensures there exists a linear prolongation operator E which projects $H^{2\alpha}(\Omega)$ onto $H^{2\alpha}(\mathbb{R}^2)$.

4.3 Fractional Order Anisotropic Diffusion

We used anisotropic diffusion in the case of PdEs model to reduce noise while preserving edges, but because of the uniform treatment and less of PdEs model, it was less efficient at the noise reduction vs edge prevention trade-off.

Let t denote time and c the diffusion coefficient, then anisotropic diffusion may be formulated as

$$\frac{\partial u}{\partial t} = \operatorname{div}(c(|\nabla u|^2)\nabla u) \tag{4.5}$$

This equation is associated with the following energy functional

$$E(u) = \int_{\Omega} f(|\nabla u|) d\Omega \tag{4.6}$$

where Ω is the image support, and f is an increasing function associated with diffusion coefficient as

$$c(s) = \frac{f'(\sqrt{s})}{\sqrt{s}}$$

Similarly we can define fractional order anisotropic diffusion as minimizing the energy functional

$$E(u) = \int_\Omega f(|D_\alpha u|) d\Omega \tag{4.7}$$

where $|D_\alpha u| = \sqrt{D_{\alpha x}^2 + D_{\alpha y}^2}$.

We can now solve the Euler-Lagrange equation for this minimization problem. We consider a general test function $\eta \in C^\infty(\Omega)$. Define

$$\Phi(a) = \int_\Omega f\left(|D_\alpha u + a D_\alpha \eta|\right) dxdy.$$

We want to obtain the condition so that a local minimum occurs at $a = 0$ regardless of what is η. We have,

$$\begin{aligned}
\Phi'(0) &= \frac{d}{da} \int_\Omega f\left(|D_\alpha u + a D_\alpha \eta|\right) dxdy \quad \Big|_{a=0} \\
&= \int_\Omega \Bigg(f'\left(|D_\alpha u|\right) \frac{D_{\alpha x} u}{|D_\alpha u|} D_{\alpha x} \eta \\
&\quad + f'\left(|D_\alpha u|\right) \frac{D_{\alpha y} u}{|D_\alpha u|} D_{\alpha y} \eta \Bigg) dxdy \\
&= \int_\Omega \Big(D_{\alpha x}^* \left(c\left(|D_\alpha u|^2\right) D_{\alpha x} u \right) \\
&\quad + D_{\alpha y}^* \left(c\left(|D_\alpha u|^2\right) D_{\alpha y} u \right) \Big) \eta dxdy
\end{aligned}$$

for all $\eta \in C^\infty(\Omega)$, where $D_{\alpha x}^*$ is the adjoint of $D_{\alpha x}$ and $D_{\alpha y}^*$ is the adjoint of $D_{\alpha y}$. Thus, the Euler-Lagrange equation is

$$D_{\alpha x}^* \left(c\left(|D_\alpha u|^2\right) D_{\alpha x} u \right) + D_{\alpha y}^* \left(c\left(|D_\alpha u|^2\right) D_{\alpha y} u \right) = 0. \tag{4.8}$$

This Euler-Lagrange equation leads us to the following gradient descent procedure

$$\frac{\partial u}{\partial t} = -D_{\alpha x}^* \left(c\left(|D_{\alpha x} u|^2\right) D_{\alpha x} u \right) - D_{\alpha y}^* \left(c\left(|D_{\alpha y} u|^2\right) D_{\alpha y} u \right) \tag{4.9}$$

Chapter 5

Numerical Algorithm for Fractional Order Anisotropic Diffusion

5.1 Descretization of Fractional Order Anisotropic Diffusion

To implement the gradient descent procedure we need a discretized version of equation (4.9). Consider the image u is $m \times m$ pixels. We will utilize 2-D discrete Fourier transform (2-D DFT) given by:

$$\hat{u}(w_1, w_2) = \frac{1}{m} \sum_{x,y=0}^{m-1} u(x, y) e^{\left(-j2\pi\left(\frac{w_1 x + w_2 y}{m}\right)\right)} \tag{5.1}$$

From equation (5.1), the equivalent of first-order partial difference along x-direction in the frequency domain is

$$D_x u(x, y) = u(x, y) - u(x-1, y) \overset{F}{\leftrightarrow} \left(1 - e^{\frac{-j2\pi w_1}{m}}\right) \hat{u}(w_1, w_2) \tag{5.2}$$

Using equation (5.2) we now define the fractional-order partial difference equation along x-direction in frequency domain as

$$D_{\alpha x} u(x, y) = u(x, y) - u(x-1, y) \overset{F}{\leftrightarrow} \left(1 - e^{\frac{-j2\pi w_1}{m}}\right)^{\alpha} \hat{u}(w_1, w_2) \tag{5.3}$$

For practical computation we will use central difference for better accuracy as it takes information symmetrically from both sides of the point, it is given as

$$\tilde{D}_{\alpha x} u(x, y) = u(x + \frac{1}{2}, y) - u(x - \frac{1}{2}, y) \overset{F}{\leftrightarrow} \left(e^{\frac{j\pi w_1}{m}} - e^{\frac{-j\pi w_1}{m}}\right)^{\alpha} \hat{u}(w_1, w_2) \tag{5.4}$$

Similarly,

$$\tilde{D}_{\alpha y}u(x,y) \overset{F}{\leftrightarrow} \left(e^{\frac{j\pi w_2}{m}} - e^{\frac{-j\pi w_2}{m}}\right)^{\alpha} \hat{u}(w_1,w_2) \tag{5.5}$$

Proposition 2: When m is odd, $\tilde{D}_{\alpha x}u$ is real-valued.

Proof: We are working with shifted frequency domain to centralize zero frequency for ease of computation and verification of conjugate-symmetry. It's clear that $\hat{u}(w_1,w_2)$ is conjugate-symmetric. Now let,

$$p(w_1) = \left(e^{\frac{j\pi w_1}{m}} - e^{\frac{-j\pi w_1}{m}}\right)^{\alpha}, \quad -\frac{m}{2} \leq w_1 \leq \frac{m}{2} - 1$$

Then we have,

$$\overline{p(-w_1)} = \left(e^{\frac{j\pi w_1}{m}} - e^{\frac{-j\pi w_1}{m}}\right)^{\alpha} = p(w_1)$$

So, the function $p(w_1)$ is conjugate-symmetric with respect to $w_1 = 0$. Thus, the matrix $p\left(w_1\right)\hat{u}\left(w_1,w_2\right)\left(-m/2 \leq w_1,w_2 \leq m/2-1\right)$ is conjugate-symmetric about $w_1, W_2 = 0$, so $\tilde{D}_{\alpha x}u$ is real-valued.

When m is an even integer, then the frequency domain definition of $p(w_1)$ is not symmetric about $w_1 = 0$ and so there is complex component in $\tilde{D}_{\alpha x}u$. However, when m is large enough, the complex component is small enough to ignore the complex component. Now we define the following operators

$$K_x(\hat{u}(w_1,w_2)) \coloneqq \left(e^{\frac{j\pi w_1}{m}} - e^{\frac{-j\pi w_1}{m}}\right)^{\alpha} \hat{u}(w_1,w_2),$$
$$K_y(\hat{u}(w_1,w_2)) \coloneqq \left(e^{\frac{j\pi w_2}{m}} - e^{\frac{-j\pi w_2}{m}}\right)^{\alpha} \hat{u}(w_1,w_2)$$

where, $-\frac{m}{2} \leq w_1, w_2 \leq \frac{m}{2} - 1$. Then we have,

$$K_x^*(\hat{u}(w_1,w_2)) = conj\left(\left(e^{\frac{j\pi w_1}{m}} - e^{\frac{-j\pi w_1}{m}}\right)^{\alpha}\right)\hat{u}(w_1,w_2),$$
$$K_y^*(\hat{u}(w_1,w_2)) = conj\left(\left(e^{\frac{j\pi w_2}{m}} - e^{\frac{-j\pi w_2}{m}}\right)^{\alpha}\right)\hat{u}(w_1,w_2)$$

where, K_x^* and K_y^* are the adjoints of K_x and K_y, respectively.

We now rewrite the operator $\tilde{D}_{\alpha x}$ as,

$$\tilde{D}_{\alpha x} = F^{-1} \circ K_x \circ F$$
$$\implies \tilde{D}_{\alpha x}^* = (F^{-1} \circ K_x \circ F)^* = F^* \circ K_x^* \circ (F^{-1})^*$$
$$\implies \tilde{D}_{\alpha x}^* = F^{-1} \circ K_x^* \circ F$$

Thus,

$$\tilde{D}^*_{\alpha x} u(x,y) \overset{F}{\leftrightarrow} (K^*_x \circ F) u(x,y) \tag{5.6}$$

Similarly,

$$\tilde{D}^*_{\alpha y} u(x,y) \overset{F}{\leftrightarrow} (K^*_y \circ F) u(x,y) \tag{5.7}$$

Let

$$g = \tilde{D}^*_{\alpha x}\left(c\left(\left|\tilde{D}_\alpha u\right|^2\right)\tilde{D}_{\alpha x} u\right) + \tilde{D}^*_{\alpha y}\left(c\left(\left|\tilde{D}_\alpha u\right|^2\right)\tilde{D}_{\alpha y} u\right) \tag{5.8}$$

Then, from equation (5.6) and equation (5.7) we have,

$$\hat{g} = K^*_x \circ F\left(c\left(\left|\tilde{D}_\alpha u\right|^2\right)\tilde{D}_{\alpha x} u\right) + K^*_y \circ F\left(c\left(\left|\tilde{D}_\alpha u\right|^2\right)\tilde{D}_{\alpha y} u\right) \tag{5.9}$$

5.2 Iterative Procedure

Now we can compute the evolution of the initial image u, along the flow in equation (4.9), modify the the matrix in Fourier domain, and return to spatial domain in the end to obtain the denoised image. We follow the following steps to denoise the image:

1. Set $u_1 = u$ as the initial noisy image, and set the value of Δt.
2. Compute the 2D-DFT $\hat{u_1}$.
3. Compute the α-order partial differences using equation (5.3) and equation (5.4).
4. Compute $g_{xn} = c\left(\left|\tilde{D}_\alpha u\right|^2\right)\tilde{D}_{\alpha x} u$ and $g_{yn} = c\left(\left|\tilde{D}_\alpha u\right|^2\right)\tilde{D}_{\alpha y} u$.
5. Compute $\hat{g}_n$ using equation (5.9).
6. Compute $\hat{u}_{n+1} = \hat{u}_n - \hat{g}_n \times \Delta t$, and increment n.
7. Compute the image by $u_{n+1} = \Re(F^{-1}(\hat{u}_{n+1}))$.
8. Check if the stopping condition is met. If yes, stop. Else, go back to step 3.

During are computations, since m will be large enough, we will only consider the real part while computing the image as the complex part will be small enough to neglect.

5.3 Stability Analysis

We have the diffusion coefficient,

$$c(|\tilde{D}_\alpha u|^2) = \frac{1}{1+|\tilde{D}_\alpha u|^2} < 1$$

While analysing the stability, we consider the diffusion coefficient as a fixed constant less than 1 for efficiency, since the diffusion coefficient function is bounded above by 1.

Let $v(w) = (1-\exp(-j2\pi w/m))^\alpha \exp(j\pi w/m)$, we have

$$\begin{aligned}
&\left\|u_{n+1}(x,y)\right\|_2 = \left\|\hat{u}_{n+1}\left(w_1,w_2\right)\right\|_2 \\
&= \| \left(1-\Delta t\cdot \text{conj}\left(v\left(w_1\right)\right)\cdot c\cdot v\left(w_1\right)\ -\Delta t\cdot \text{conj}\left(v\left(w_2\right)\right)\cdot c\cdot v\left(w_2\right)\right)\hat{u}_n\left(w_1,w_2\right)\|_2 \\
&= \| \left(1-\Delta t\cdot c\cdot \left|v\left(w_1\right)\right|^2\ -\Delta t\cdot c\cdot \left|v\left(w_2\right)\right|^2\right)\hat{u}_n\left(w_1,w_2\right)\|_2 \\
&= \| \left(1-\Delta t\cdot c\cdot \left|1-\exp\left(-j2\pi w_1/m\right)\right|^{2\alpha}\ -\Delta t\cdot c\cdot \left|1-\exp\left(-j2\pi w_2/m\right)\right|^{2\alpha}\right)\hat{u}_n\left(w_1,w_2\right)\|_2 \\
&= \| \left(1-\Delta t\cdot c\cdot \left(4\sin^2\left(-\pi w_1/m\right)\right)^\alpha\ -\Delta t\cdot c\cdot \left(4\sin^2\left(-\pi w_2/m\right)\right)^\alpha\right)\hat{u}_n\left(w_1,w_2\right)\|_2
\end{aligned}$$

where $\|\cdot\|_2$ denotes the ℓ^2 norm.

So, the stability condition will be satisfied when

$$\frac{1-\Delta t\cdot c\cdot(4\sin^2(-\pi w_1/m))^\alpha}{1-\Delta t\cdot c\cdot(4\sin^2(-\pi w_2/m))^\alpha} \le 1$$

for all $-\frac{m}{2} \le w_1, w_2 \le \frac{m}{2}-1$.

Thus, $\Delta t\cdot c \le 4^{-\alpha}$. Since $c<1$, we have the stability condition $\Delta t \le 4^{-\alpha}$ in ℓ^2 norm.

Chapter 6

Results

6.1 Results of PdE Model

From equation (3.13), we have the following scheme for diffusion on graph,

$$f^{n+1}(u) = f^n(u) + \lambda\alpha\Delta_{w,\infty}f^n(u) + \lambda\beta\Delta_{w,2}f^n(u)$$

For our use case of image denoising, the initial value $f^0 : V \to \mathbb{R}$ will be the corresponding pixel intensity values of the input image. The weight function is governed by two similarity functions in the various cases shown in Figure 6.1:

$$g_0(u,v) = 1$$
$$g_1(u,v) = e^{\left(-\mu(u,v)/(0.4)^2\right)}$$

where $\mu(u,v)$ is the euclidean distance between the patch feature vectors in the case of non-local graph, and absolute difference of pixel values in the case of local graph.
The first row shows results with $\alpha = 1$ and $\beta = 0$ (∞-Laplacian) following the scheme:

$$f^{n+1}(u) = f^n(u) + \lambda\Delta_{w,\infty}f^n(u)$$

Second row shows results with $\alpha = 0.5$ and $\beta = 0.5$ (an intermediate) following the scheme:

$$f^{n+1}(u) = f^n(u) + \frac{\lambda}{2}\left[\Delta_{w,\infty}f^n(u) + \Delta_{w,2}f^n(u)\right]$$

The third row shows results with $\alpha = 0$ and $\beta = 1$ (Laplacian) following the scheme:

$$f^{n+1}(u) = f^n(u) + \lambda\Delta_{w,2}f^n(u)$$

First and fourth columns show results computed for a 4-adjacency grid graph with a constant weight function (g_0). Second and fifth columns show results computed on a 4-adjacency grid graph with a image dependent weight function (g_1).
Third and sixth columns show results computed on a non-local grid graph, using a 15×15 neighborhood window and a 5×5 patch as feature vector.

As evident from Figure 6.1, the non-local graph with feature dependent weight function achieves better denoising with better edge preservation than the other configurations. And using the parameters α and β we can now have more control on the processed images for any particular input image than other previous methods.

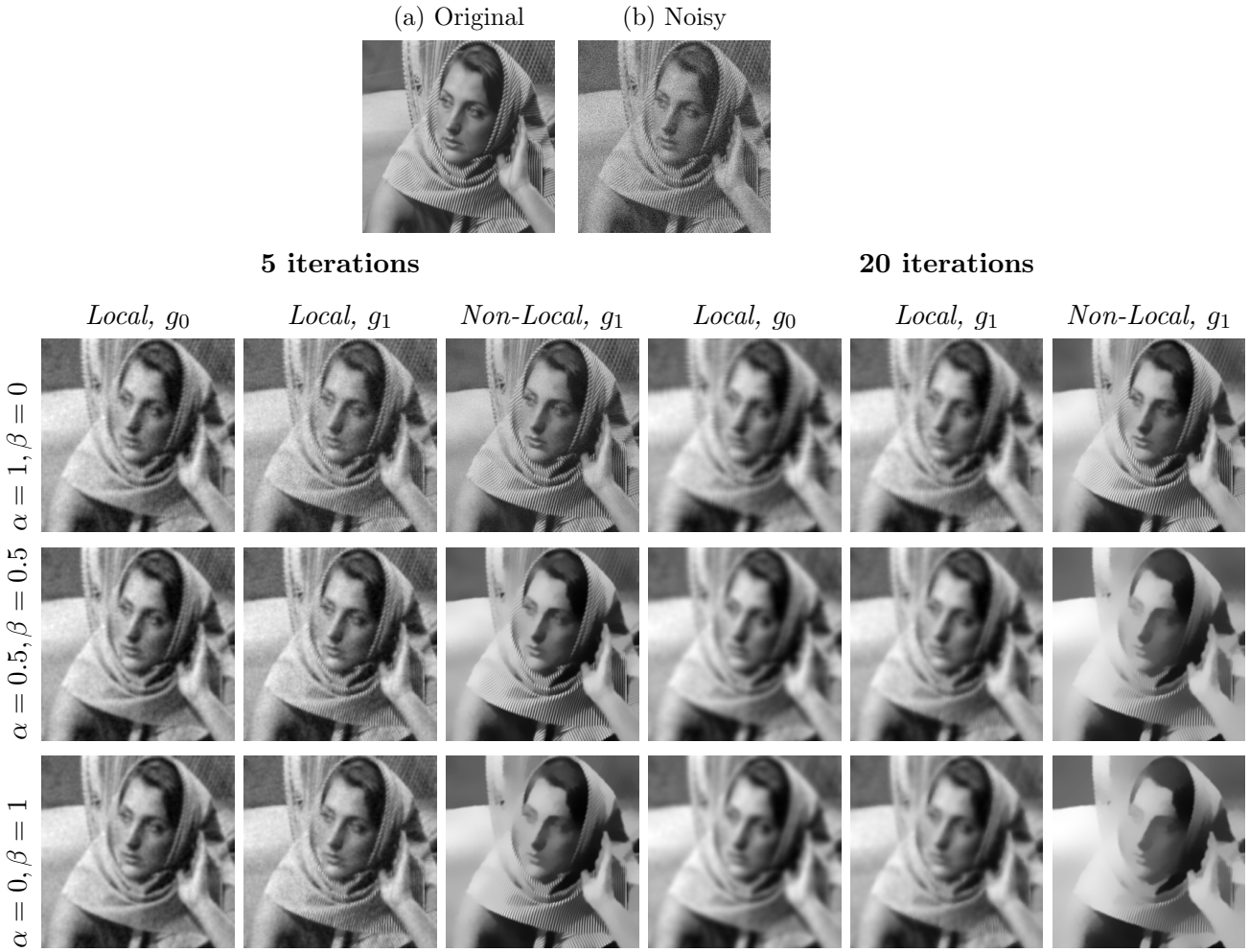


Figure 6.1: Image denoising of a gray scale image with different parameter and graph configurations. See the text above for details.

6.2 Results of FPdE Model

Here we present the results of image denoising based on the algorithm presented in section 5.2. For uniformity, we tested the algorithm on the image "Barbara" with size 278×278. We have the results for different values of α from $\alpha = 1.0$ to $\alpha = 2.2$. We are using the following function for diffusion coefficient:

$$c(s) = \frac{1}{1+s}$$

As shown in section 5.3, the difference scheme is stable for $\Delta t \leq 4^{-\alpha}$. We have taken $\Delta t = 0.01$ through out our experiment.



(a) $\alpha = 1.0$ (b) $\alpha = 1.1$ (c) $\alpha = 1.2$ (d) $\alpha = 1.3$ (e) $\alpha = 1.4$



(f) $\alpha = 1.5$ (g) $\alpha = 1.6$ (h) $\alpha = 1.7$ (i) $\alpha = 1.8$ (j) $\alpha = 1.9$



(k) $\alpha = 2.0$ (l) $\alpha = 2.1$ (m) $\alpha = 2.2$

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